DESIGN COMPUTED · FREE GUIDE

Borrow a tool: how aerospace software made a crumpled-paper building

The free guide to the reel on the Guggenheim Bilbao: how Frank Gehry's team turned paper models into buildable panels with software from the aerospace industry, and how you can borrow tools the same way.



How it works

Frank Gehry designs with physical models: crumpled paper and card. The shape is too complex to draw with ordinary plans. His team measured the model with a digitiser (points in 3D), then let software turn those points into smooth surfaces.

The software was CATIA, made by Dassault Systèmes for the French aerospace industry. Once the building was a digital model, it could be cut into thousands of parts, each with exact dimensions for fabrication, and the same files went to the people who made and assembled the pieces.

The method is one most designers can copy: measure a hand-made model, build a clean surface from the points, then split that surface into panels a factory can make.

The Guggenheim Bilbao in facts

The museum opened in 1997. Its skin is covered in thin titanium panels (sources give different counts, so treat any single number with care). Gehry's team first used CATIA on the Barcelona fish sculpture, then took the method to the museum, one of the first major buildings designed this way.

Gehry Technologies later adapted CATIA into Digital Project, software made for architecture.

Built examples

Guggenheim Museum Bilbao (1997)	Frank Gehry: paper models digitised, surfaces built in CATIA, titanium skin cut into panels.
Walt Disney Concert Hall, Los Angeles (2003)	Gehry's office: the same digital workflow for a complex stainless-steel skin.
Fondation Louis Vuitton, Paris (2014)	Gehry's office: glass sails modelled and fabricated from digital files.
Digital Project	Software built on CATIA V5 for architects, developed by Gehry Technologies, later owned by Trimble.

Try it yourself

1. Crumple a sheet of paper, then relax it slightly to get a soft, wild shape.
2. Photograph it from several sides with a coin next to it for scale.
3. Turn the photos into a 3D model with a free photogrammetry tool such as Meshroom, or trace the points by hand in Blender.
4. Smooth the result into a surface (in Blender: Remesh, then Subdivision Surface).
5. Cut the surface into panels with a grid, and number every panel.
6. Print or cut the panels from paper and rebuild the shape on a table: where must the panels bend, and where can they stay flat?

The rule to remember

When a shape won't draw, borrow a tool from another industry.

Digitising without checking	Always compare the digital surface with the physical model.
Smoothing too much	You lose the detail that made the shape interesting.
Forgetting the factory	Panel size and bending limits decide what can really be built.

Go further

Frank Gehry and the digital breakthrough behind the Guggenheim Bilbao, Dassault Systèmes
https://blog.3ds.com/industries/architecture-engineering-construction/frank-gehry-and-the-digital-breakthrough-that-made-the-guggenheim-bilbao-possible/

Digital Project, Wikipedia
https://en.wikipedia.org/wiki/Digital_Project

Gehry Technologies' lasting influence, Bluebeam
https://blog.bluebeam.com/gehry-technologies-industry-influence/

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Facts: Dassault Systèmes, Wikipedia, Bluebeam, LACMA.

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